RF Sim for KiCad – Worked Example

An 868 MHz sensor board, from the layout to the matching values

Version 0.27.2 · November 2026

This guide follows one design from the layout to the matching values. Every step shows the window as you will see it (the screenshots were taken with release 0.26.4; the tuning viewer of later versions also offers *fitted as C* / *fitted as L*). The *User Manual* explains each page and option in full; the *Installation Guide* describes the setup of KiCad, the plugin and openEMS.

1 The board

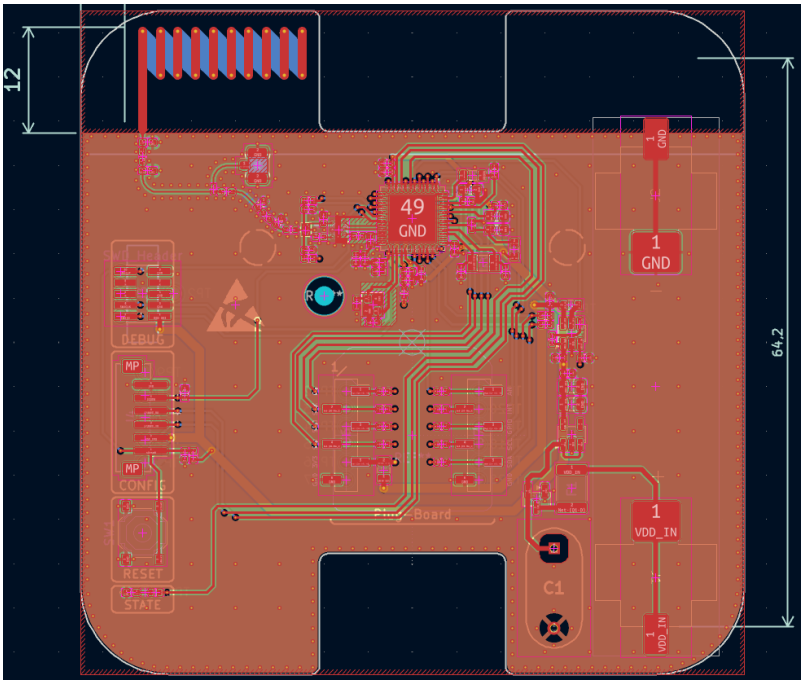


Figure 1: The example: a room sensor with an STM32WLE5 radio and a printed helix antenna (top left, 12 mm high) above a cut-out in the ground plane.

The radio works in the 868 / 915 MHz bands. Its RF output runs along the left edge to the antenna. On the way it passes the test connector J5 and the matching network:

- Z1: capacitor to ground, 5.1 pF;
- Z2: series inductor, 3.3 nH;
- Z3: capacitor to ground on the antenna side, 3.3 pF;
- L5: an optional series place, not fitted.

Two questions: how good is the antenna on this board, and are these values right for it?

2 Step 1: select the RF chain in the PCB editor

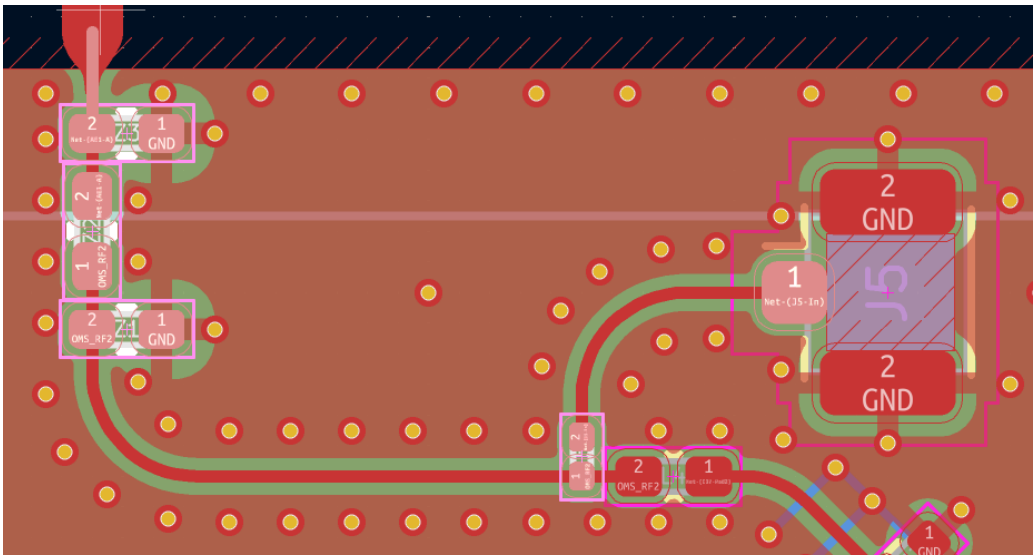


Figure 2: The RF chain in the PCB editor: J5 (right), L5, the trace to the matching network Z1 – Z2 – Z3 (left) and the start of the antenna trace (top).

Open the board and press **RF Sim**. In the PCB editor select the parts you want to tune: Z1, Z2, Z3 and L5. Also select one segment of the antenna trace right after Z3. This tells RF Sim where the antenna starts.

3 Step 2: create the setups

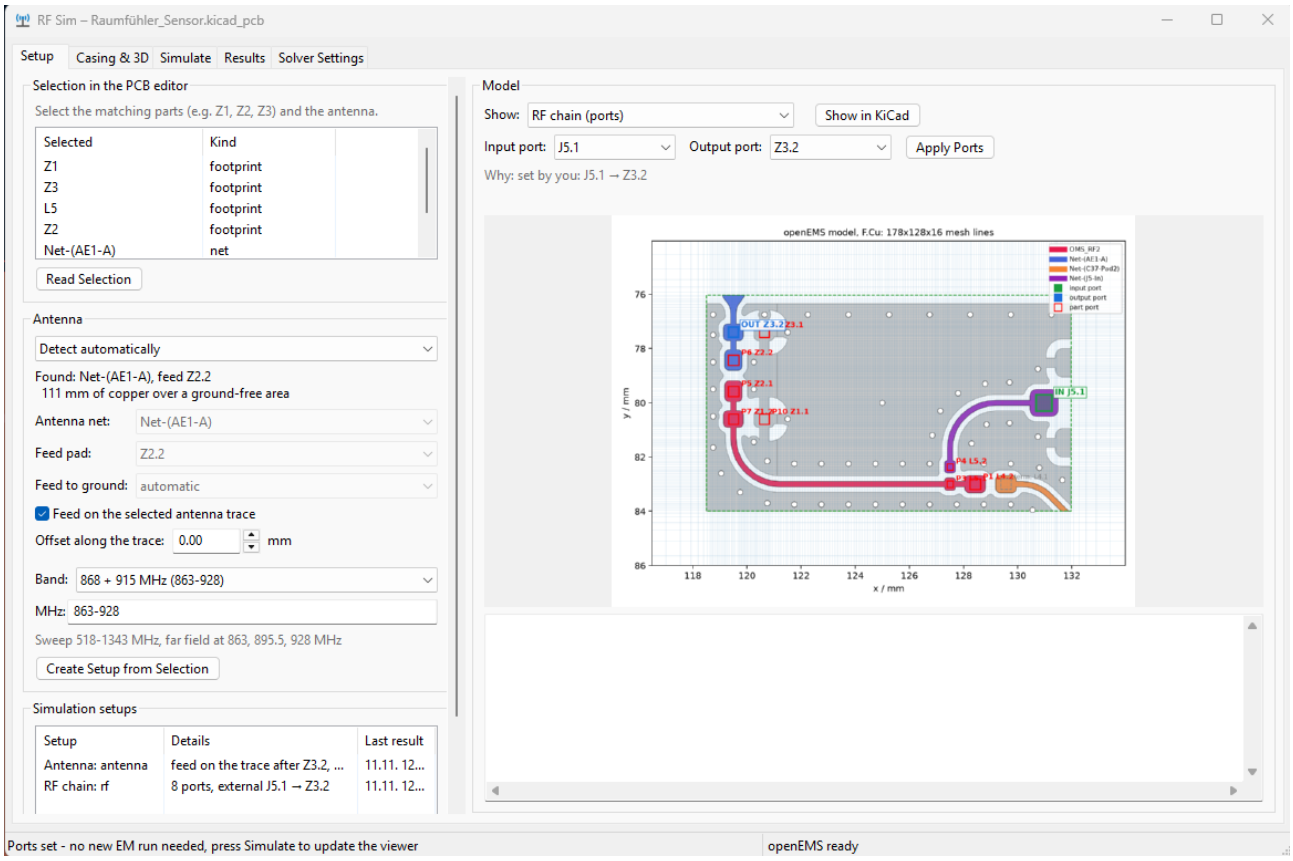


Figure 3: Setup page after **Read Selection** and **Create Setup from Selection**: the RF chain model with the input port J5.1 (green), the output port Z3.2 (blue) and the part ports (red).

1. Press **Read Selection**. The list shows Z1, Z3, L5, Z2 and the antenna net *Net-(AE1-A)*.

2. **Antenna**: *Detect automatically* finds the antenna: *Net-(AE1-A)*, 111 mm of copper over a ground-free area.
3. Tick **Feed on the selected antenna trace**. The feed then sits on the antenna trace right after the last matching pad Z3.2, where the antenna starts. **Offset along the trace** stays 0.
4. **Band**: *868 + 915 MHz (863-928)*. RF Sim derives the sweep (518–1 343 MHz) and the far-field frequencies (863, 895.5 and 928 MHz) from it.
5. Press **Create Setup from Selection**. Two setups appear under **Simulation setups**: *Antenna* and *RF chain*.

RF Sim chooses the ports itself. Here the test connector J5 was set as the input port by hand (**Input port** *J5.1*, **Apply Ports**), so the result compares directly with a measurement at J5. **Why** below the port fields shows the reason: *set by you: J5.1 → Z3.2*. Changing the ports needs no new simulation.

4 Step 3: choose the solver settings

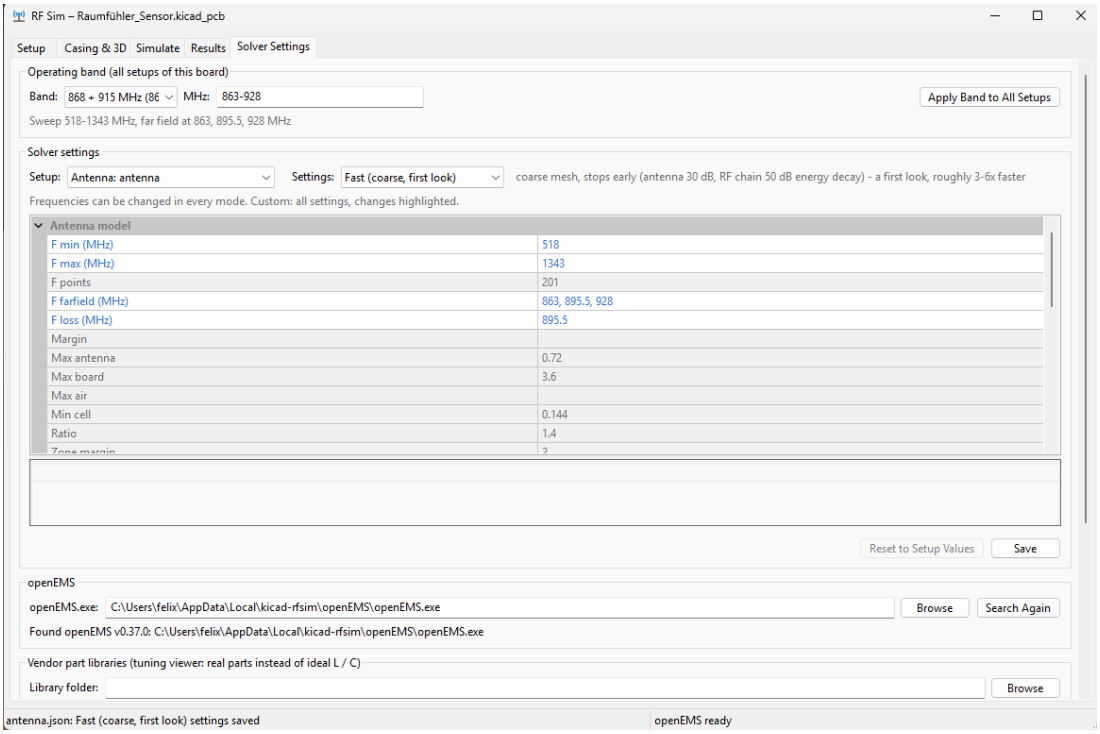


Figure 4: Solver Settings: operating band, the preset per setup and openEMS.

On **Solver Settings** choose the preset per setup. Start with *Fast* for a first look: a coarse mesh that stops early, about 3–6 times faster than *Automatic*. Use *Automatic* or *Accurate* for the final numbers. Check that openEMS is found (*Found openEMS v0.37.0*).

5 Step 4: inspect the mesh

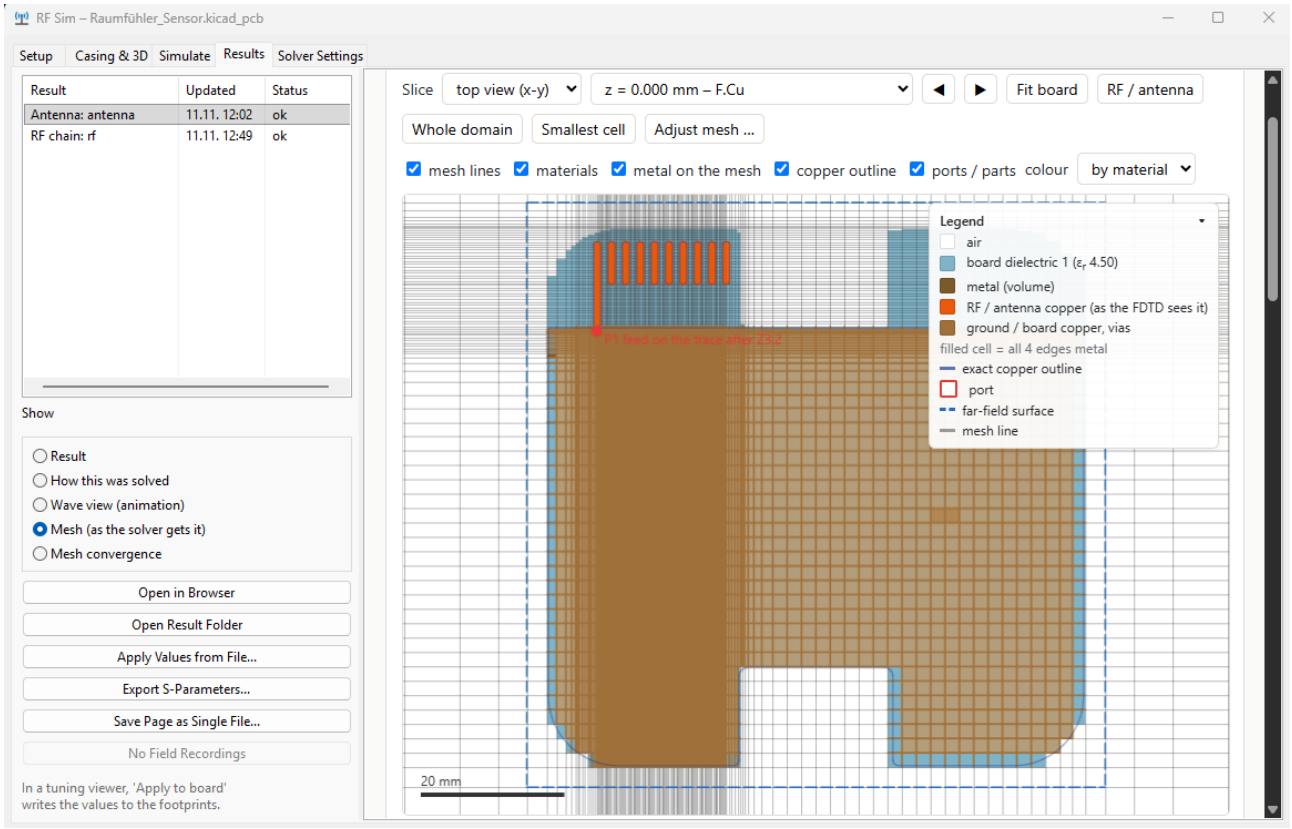


Figure 5: The antenna model on its mesh: the helix and the board copper as the solver sees them, the feed port P1 on the trace after Z3.2, and the far-field box (dashed).

Inspect Mesh on the Setup page builds the model in a few seconds, without a simulation. The mesh opens on **Results** with **Show**: *Mesh (as the solver gets it)*. Check three things:

- every turn of the antenna is in the model and does not touch the next one;
- the feed port sits where the antenna starts;
- the smallest cell (button **Smallest cell**) lies on a real detail, not on a sliver of copper.

6 Step 5: simulate

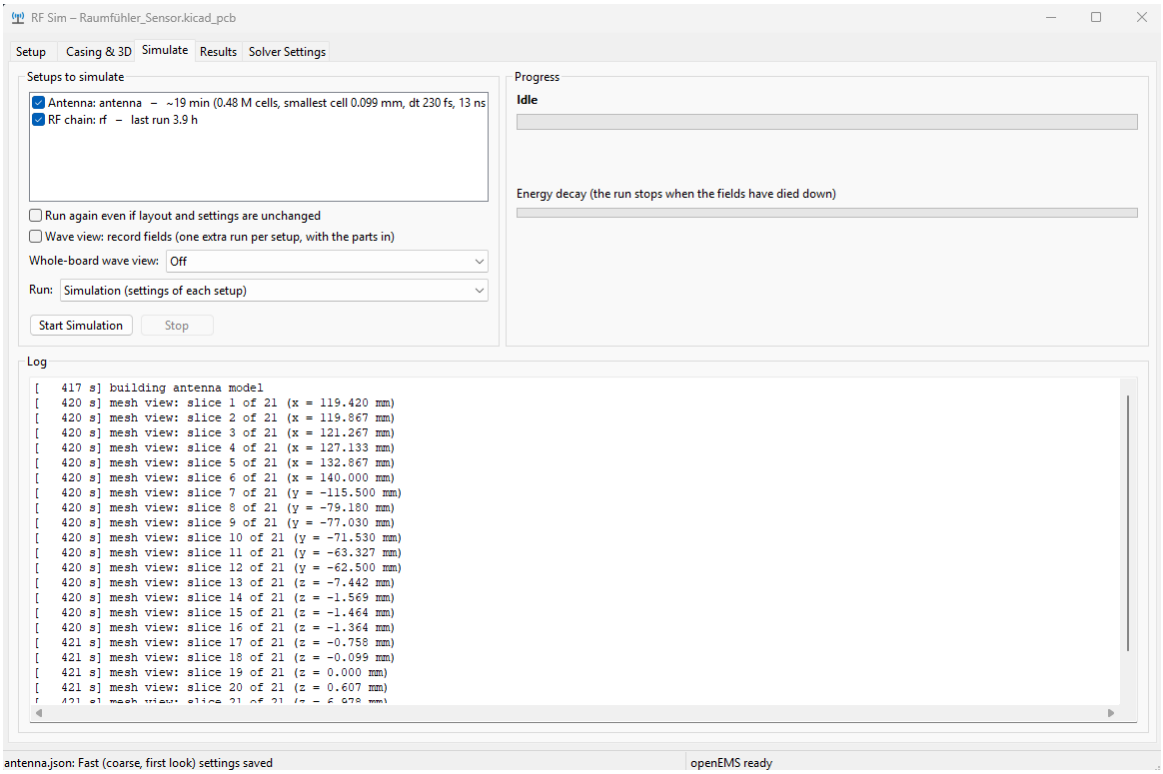


Figure 6: Simulate page: the estimate per setup, the options and the log.

On **Simulate** tick both setups and press **Start Simulation**. The list shows an estimate per setup: here about 19 min for the antenna on *Fast* (0.48 M cells). The estimate is cautious: the real run usually ends sooner, because it stops when the fields have died down. The RF chain of this board took 3.9 h on *Automatic* on an older 4-core PC with release 0.26.4, which needed one run per pad port. Since 0.27 a tuned part has one port across its pads, and the excitation pulse is shorter; both shorten the RF chain.

Simulation time. openEMS runs on the CPU only. Plan minutes for a first look and hours for final numbers. Simulate overnight, and keep the selection small: every selected part adds runs to the RF chain.

7 Step 6: read the antenna result

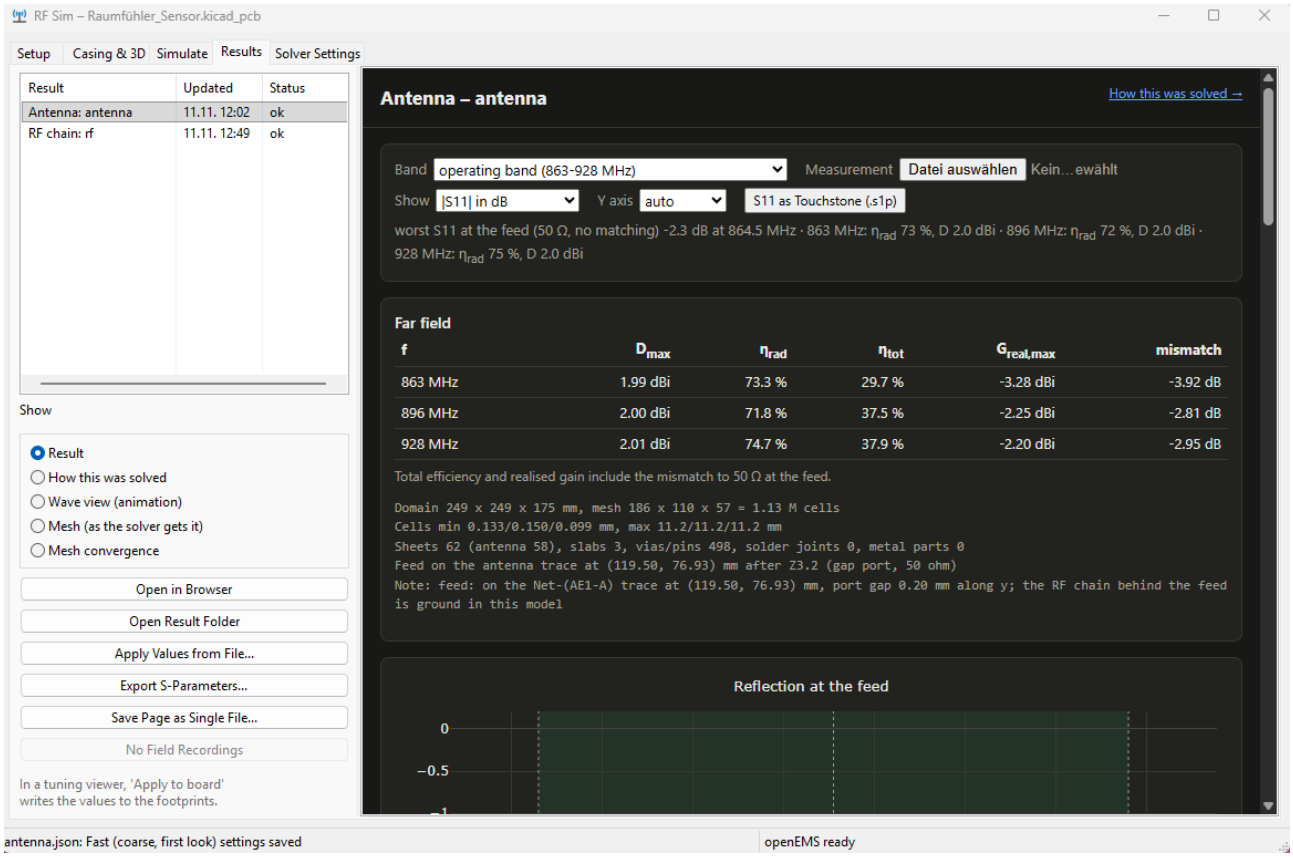


Figure 7: Antenna report: band, reflection at the feed and the far-field table.

Results → *Antenna*. The report shows the antenna as fed directly with 50 Ω, without the matching network:

- radiation efficiency 72–75 % and directivity 2.0 dBi over the band: a good antenna for its size;
- total efficiency only 30–38 %: the mismatch to 50 Ω costs 2.8 dB to 3.9 dB. This is what the matching network has to recover;
- the S11 curve and the Smith chart below the table show where the antenna resonates. **Measurement** loads a measured `.s1p` file into the same chart.

The feed note says *the RF chain behind the feed is ground in this model*: the antenna setup sees the board up to the feed. The matching network is the job of the RF chain.

8 Step 7: tune the matching network

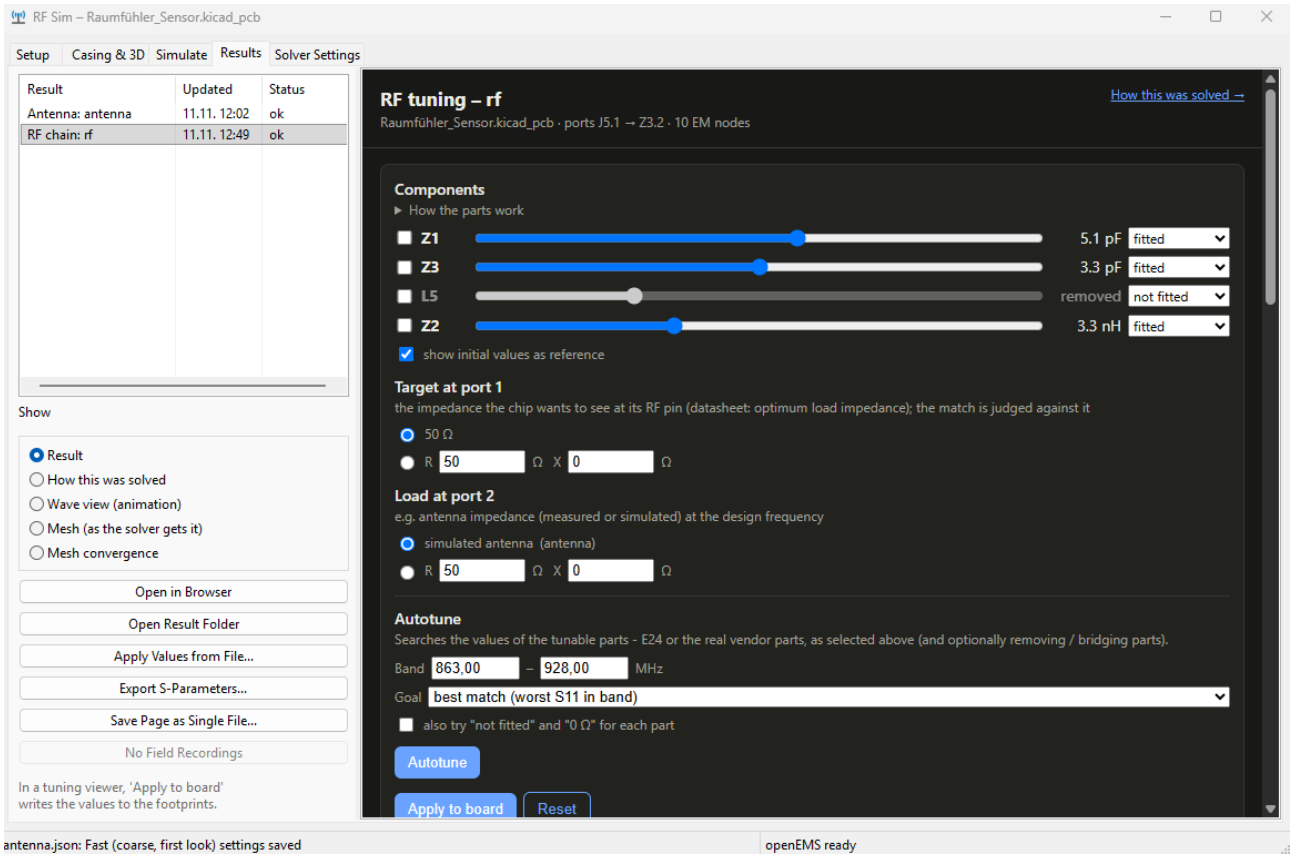


Figure 8: Tuning viewer of the RF chain: one slider per part, the target at port 1, the load at port 2 and the autotune.

Results → *RF chain*. The tuning viewer combines the simulated layout with the simulated antenna:

- **Load at port 2**: *simulated antenna*. The values are tuned for this antenna on this board.
- **Target at port 1**: 50 Ω for a test connector or a 50 Ω radio. For a chip with another optimum load, enter R and X from its datasheet.
- Move a slider: S11, the Smith chart and the power to the antenna follow at once. The selector next to each part sets *fitted as C*, *fitted as L*, *not fitted* or *0 Ω*.
- **Autotune**: set the band (863–928 MHz) and the goal (best worst-case S11 in the band). **C / L**: keep each part as set, or let the autotune choose C or L per part. Tick **also try "not fitted" and "0 Ω"** to let it remove parts.

9 Step 8: write the values to the board

Press **Apply to board** and confirm. RF Sim writes the values to the footprints in one undo step. Then run **Tools** → **Update Schematic from PCB** in KiCad. Simulate the RF chain once more with the new values for a final check, and measure the board: the plugin gives a good starting point, not the last word.

> **Try it yourself.** The boards pack (`kicad-rfsim-boards-<version>.zip`) contains the boards of a public antenna kit with ready setups. Board `ti_dk2/ti06` is a similar 868 MHz helix with a three-part matching network: open it and repeat steps 3 to 7.